

Vector Functions

Derivative: If $\mathbf{r}(t)$ is a vector-valued function, we define the **derivative of $\mathbf{r}$ with respect to t** to be the vector-valued function $\mathbf{r}'$ given by

$$\mathbf{r}' = \lim_{h \rightarrow 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}$$

The domain of $\mathbf{r}$ consists of all values of t in the domain of $\mathbf{r}(t)$ for which the limit exists.

Rules of differentiation: Let $\mathbf{r}(t)$, $\mathbf{r}_1(t)$ and $\mathbf{r}_2(t)$ be differentiable vector-valued functions that are all in 2-space or all in 3-space, and let $f(t)$ be a differentiable real-valued function, k a scalar, and $\mathbf{c}$ a constant vector (that is, a vector whose value does not depend on t). Then the following rules of differentiation hold:

- $\frac{d}{dt}(\mathbf{c}) = \mathbf{0}$
- $\frac{d}{dt}(k\mathbf{r}(t)) = k \frac{d}{dt}(\mathbf{r}(t))$
- $\frac{d}{dt}(k\mathbf{r}(t)) = k \frac{d}{dt}(\mathbf{r}(t))$
- $\frac{d}{dt}[\mathbf{r}_1(t) + \mathbf{r}_2(t)] = \frac{d}{dt}[\mathbf{r}_1(t)] + \frac{d}{dt}[\mathbf{r}_2(t)]$
- $\frac{d}{dt}[\mathbf{r}_1(t) - \mathbf{r}_2(t)] = \frac{d}{dt}[\mathbf{r}_1(t)] - \frac{d}{dt}[\mathbf{r}_2(t)]$
- $\frac{d}{dt}(f(t)\mathbf{r}(t)) = f(t) \frac{d}{dt}(\mathbf{r}(t)) + \mathbf{r}(t) \frac{d}{dt}[f(t)]$

Example-1: Let $\mathbf{r}(t) = t^2\mathbf{i} + e^t\mathbf{j} - 2\cos\pi t\mathbf{k}$. Find $\mathbf{r}'(t)$.

Solution: $\mathbf{r}'(t) = \frac{d}{dt}[t^2\mathbf{i} + e^t\mathbf{j} - 2\cos\pi t\mathbf{k}]$

$$= \frac{d}{dt}(t^2)\mathbf{i} + \frac{d}{dt}(e^t)\mathbf{j} - \frac{d}{dt}(2\cos\pi t)\mathbf{k}$$

$$= 2t\mathbf{i} + e^t\mathbf{j} + 2\pi \sin\pi t\mathbf{k}$$

Integration of vector valued function: Let $\mathbf{r}(t) = x\mathbf{i} + y\mathbf{j}$

If $\mathbf{r}(t)$ be a vector valued function that is continuous in the interval $a \leq t \leq b$ then the definite integral of $\mathbf{r}(t)$,

$$\int_a^b \mathbf{r}(t) dt = \left(\int_a^b x(t) dt \right) \mathbf{i} + \left(\int_a^b y(t) dt \right) \mathbf{j} \quad [\text{In 2-space}]$$

$$\int_a^b \mathbf{r}(t) dt = \left(\int_a^b x(t) dt \right) i + \left(\int_a^b y(t) dt \right) j + \left(\int_a^b z(t) dt \right) k \quad [\text{In 3-space}]$$

Example-2: Let $\mathbf{r}(t) = t^2 i + e^t j - 2 \cos \pi t k$. Find $\int_0^1 \mathbf{r}(t) dt$.

Solution: $\int_0^1 \mathbf{r}(t) dt = \left(\int_0^1 t^2 dt \right) i + \left(\int_0^1 e^t dt \right) j - 2 \left(\int_0^1 \cos \pi t dt \right) k$

$$= \frac{t^3}{3} \Big|_0^1 + e^t \Big|_0^1 - \frac{2}{\pi} \sin \pi t \Big|_0^1$$

$$= \frac{1}{3} i + (e - 1) j$$

Antiderivative of vector valued function:

An antiderivative for a vector-valued function $\mathbf{r}(t)$ is a vector-valued function $\mathbf{R}(t)$ such that

$$\mathbf{R}'(t) = \mathbf{r}(t)$$

In integral notation,

$$\int \mathbf{r}(t) dt = \mathbf{R}(t) + \mathbf{C}$$

where $\mathbf{C}$ represents an arbitrary constant vector.

Example-3: Evaluate $\int (2ti + 3t^2j) dt$

Solution: $\int (2ti + 3t^2j) dt = \left(\int 2t dt \right) i + \left(\int 3t^2 dt \right) j$

$$= (t^2 + c_1) i + (t^3 + c_2) j$$

$$= (t^2 i + t^3 j) + (c_1 i + c_2 j)$$

$$= (t^2 i + t^3 j) + \mathbf{C}$$

Example-4: Evaluate $\int_0^2 (2ti + 3t^2j) dt$

$$= t^2 \Big|_0^2 i + t^3 \Big|_0^2 j$$

$$= 4i + 8j$$

Example-5: Find $\mathbf{r}(t)$ where $\mathbf{r}'(t) = \langle 3, 2t \rangle$, and $\mathbf{r}(1) = \langle 2, 5 \rangle$

Solution: Integrating $\mathbf{r}'(t)$ to obtain $\mathbf{r}(t)$

$$\mathbf{r}(t) = \int \mathbf{r}'(t) dt = \int \langle 3, 2t \rangle dt = \langle 3t, t^2 \rangle + \mathbf{C}$$

Where $\mathbf{C}$ is vector constant of integration. Now substitute $t = 1$

$$\mathbf{r}(1) = \langle 3, 1 \rangle + \mathbf{C} = \langle 2, 5 \rangle$$

So that $C = \langle -1, 4 \rangle$

Thus $r(t) = \langle 3t, t^2 \rangle + \langle -1, 4 \rangle = \langle 3t - 1, t^2 + 4 \rangle$

Smooth Parametrized Curve: We will say that a curve represented by $\mathbf{r}(t)$ is **smoothly parametrized** by $\mathbf{r}(t)$, or that $\mathbf{r}(t)$ is a **smooth function** of t if $r'(t)$ is continuous and $r'(t) \neq \mathbf{0}$ for any allowable value of t . Geometrically, this means that a smoothly parametrized curve can have no abrupt changes in direction as the parameter increases.

Arc Length:

If $x = f(t), y = g(t)$

Where $a \leq t \leq b$

Then the arc length is, $L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$

Analogously

$x = x(t), y = y(t), z = z(t)$

Where $a \leq t \leq b$

Then arc length, $L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$

In vector form, $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$ or $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$

It follows that $\frac{d\mathbf{r}}{dt} = \frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j}$ or $\frac{d\mathbf{r}}{dt} = \frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j} + \frac{dz}{dt}\mathbf{k}$

And hence $\left\|\frac{d\mathbf{r}}{dt}\right\| = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$ or $\left\|\frac{d\mathbf{r}}{dt}\right\| = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}$

Theorem: If c is the graph in 2-space or 3-space of a smooth vector valued function $\mathbf{r}(t)$, then its arc length L from $t = a$ to $t = b$ is,

$$L = \int_a^b \left\|\frac{d\mathbf{r}}{dt}\right\| dt$$

Example-5: Find the arc length of that portion of the circular helix, $x = \cos t, y = \sin t, z = t$, from $t = 0$ to $t = \pi$

Solution: $\mathbf{r}(t) = \cos t \mathbf{i} + \sin t \mathbf{j} + t \mathbf{k} = \langle \cos t, \sin t, t \rangle$

$\mathbf{r}'(t) = \langle -\sin t, \cos t, 1 \rangle$

$$\left\|\frac{d\mathbf{r}}{dt}\right\| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1} = \sqrt{2}$$

Therefore arc length of the helix,

$$\begin{aligned} L &= \int_0^{\pi} \left\| \frac{dr}{dt} \right\| dt \\ &= \int_0^{\pi} \sqrt{2} dt = \sqrt{2}\pi \text{ unit} \end{aligned}$$

Example-6: Find the arc length of the curve represented by $x = \cos^3 t, y = \sin^3 t, z = 2$ for $0 \leq t \leq \pi/2$.

Solution:

$$x = \cos^3 t$$

$$\frac{dx}{dt} = -3\cos^2 t \sin t$$

$$y = \sin^3 t$$

$$\frac{dy}{dt} = 3\sin^2 t \cos t$$

$$\frac{dz}{dt} = 0$$

$$\begin{aligned} \left\| \frac{dr}{dt} \right\| &= \sqrt{(-3\cos^2 t \sin t)^2 + (3\sin^2 t \cos t)^2} \\ &= \sqrt{9\cos^4 t \sin^2 t + 9\cos^2 t \sin^4 t} \\ &= \sqrt{9\cos^2 t \sin^2 t (\cos^2 t + \sin^2 t)} \\ &= 3 \cos t \sin t \end{aligned}$$

Therefore arc length,

$$\begin{aligned} L &= \int_0^{\pi/2} \left\| \frac{dr}{dt} \right\| dt \\ &= \int_0^{\pi/2} 3 \cos t \sin t dt \\ &= \frac{3}{2} \sin^2 t \Big|_0^{\pi/2} \\ &= \frac{3}{2} \text{ unit} \end{aligned}$$

Motion in Space:

	2-space	3-space
Position	$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$	$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$
Velocity	$\mathbf{v}(t) = \frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j}$	$\mathbf{v}(t) = \frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j} + \frac{dz}{dt}\mathbf{k}$
Acceleration	$\mathbf{a}(t) = \frac{d^2x}{dt^2}\mathbf{i} + \frac{d^2y}{dt^2}\mathbf{j}$	$\mathbf{a}(t) = \frac{d^2x}{dt^2}\mathbf{i} + \frac{d^2y}{dt^2}\mathbf{j} + \frac{d^2z}{dt^2}\mathbf{k}$
Speed	$\ \mathbf{v}(t)\ = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$	$\ \mathbf{v}(t)\ = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}$

Example-7: A particle moves along a circular path in such a way that its x - and y -coordinates at time t are $x = 2 \cos t, y = 2 \sin t$

- (a) Find the instantaneous velocity and speed of the particle at time t .
(b) Show that at each instant the acceleration vector is perpendicular to the velocity vector.

Solution (a): At time t , the position vector is

$$\mathbf{r}(t) = 2\cos t \mathbf{i} + 2\sin t \mathbf{j}$$

Instantaneous velocity,

$$\begin{aligned} \mathbf{v}(t) &= \frac{d\mathbf{r}}{dt} \\ &= \frac{dx}{dt}\mathbf{i} + \frac{dy}{dt}\mathbf{j} \\ &= -2 \sin t \mathbf{i} + 2 \cos t \mathbf{j} \end{aligned}$$

$$\begin{aligned} \text{Speed, } \|\mathbf{v}(t)\| &= \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \\ &= \sqrt{(-2\sin t)^2 + (2\cos t)^2} = 2 \end{aligned}$$

$$\begin{aligned} \text{(b): Acceleration, } \mathbf{a}(t) &= \frac{d^2x}{dt^2}\mathbf{i} + \frac{d^2y}{dt^2}\mathbf{j} \\ &= -2 \cos t \mathbf{i} - 2 \sin t \mathbf{j} \end{aligned}$$

One way of showing that $\mathbf{v}(t)$ and $\mathbf{a}(t)$ are perpendicular is to show that their dot product is zero (try it). However, it is easier to observe that $\mathbf{a}(t)$ is the negative of $\mathbf{r}(t)$, which implies that $\mathbf{v}(t)$ and $\mathbf{a}(t)$ are perpendicular, since at each point on a circle the radius and tangent line are perpendicular.

Example-8: A particle moves through 3-space in such a way that the velocity is $v(t) = i + tj + t^2k$. Find the coordinates of the particle at time $t = 1$ given that the particle is at the point $(-1, 2, 4)$ at time $t = 0$.

Solution: Integrating the velocity to obtain the position vector,

$$\begin{aligned} r(t) &= \int v(t) dt = \int (i + tj + t^2k) dt \\ &= ti + \frac{t^2}{2}j + \frac{t^3}{3}k + C \end{aligned} \quad (1)$$

Where C is the vector constant of integration. Since the coordinates of the particle at time $t = 0$ is at the point $(-1, 2, 4)$, the position vector at $t = 0$ is,

$$r(t) = -i + 2j + 4k \quad (2)$$

Now substituting $t = 0$ in equation (1) and equating with equation (2) we get,

$$C = -i + 2j + 4k$$

Now substituting the value of C in equation (1) we get,

$$\begin{aligned} r(t) &= ti + \frac{t^2}{2}j + \frac{t^3}{3}k - i + 2j + 4k \\ &= (t - 1)i + \left(\frac{t^2}{2} + 2\right)j + \left(\frac{t^3}{3} + 4\right)k \end{aligned}$$

Therefore the position vector at $t = 1$

$$r(t) = 0i + \frac{5}{2}j + \frac{13}{3}k$$

So, it's coordinates at that instant $\left(0, \frac{5}{2}, \frac{13}{3}\right)$

Exercise:

1. Find $r'(t)$ where $r(t) = \tan^{-1} t i + t \cos t j - \sqrt{t} k$
2. Evaluate $\int (t^2 i - 2tj + \frac{1}{t}k) dt$
3. Evaluate $\int_0^{\pi/2} \langle \cos 2t, \sin 2t \rangle dt$
4. Find the arc length of the curve represented by $x = e^t, y = e^{-t}, z = \sqrt{2}t$ for $0 \leq t \leq 1$.
5. Find the arc length of the curve represented by $r(t) = t^2 i + (\cos t + t \sin t)j + (\sin t - t \cos t)k$ for $0 \leq t \leq \pi$.